Project echo – Sprint 2 – Dataset Cleaning, Validation and Rebalancing Support

Sprint 2 task: Dataset Cleaning, Validation and Rebalancing Support

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1. Task Overview

The purpose of this sprint was to improve the quality and consistency of the current main dataset through systematic cleaning and validation, followed by support for dataset rebalancing. The work reused the dataset-quality validation approach established in Sprint 1 and applied it to the current dataset to identify and address invalid, unreadable, duplicated, unsupported, and inconsistent records.

The cleaning process was designed as part of a continuous workflow covering **dataset validation, cleaning, rebalancing, and final verification**, rather than treating these activities as separate handoffs. This ensured that the dataset passed between stages with its quality and structure maintained.

The final objective was to produce a cleaned and validated dataset, support the creation of a balanced dataset, and verify that the resulting dataset was suitable for the next stage of the project, particularly augmentation.

2. Sprint 2 Objective

The main objective of Sprint 2 was to establish a reliable and reproducible dataset-cleaning workflow that could be applied to the current main dataset.

The sprint focused on the following requirements:

- Reuse the Sprint 1 dataset-quality validation process on the current main dataset.
- Identify and clean invalid and unreadable audio records.
- Identify and address duplicate records.
- Remove unsupported audio formats.
- Remove recordings that did not meet the minimum duration requirement.
- Standardise species labels to ensure consistent class identification.
- Verify that the cleaning decisions were correctly applied.
- Produce a cleaned dataset and supporting cleaning reports.
- Support the rebalancing of the cleaned dataset.
- Validate the resulting balanced dataset and prepare it for augmentation.

The overall workflow followed the sequence:

Main Dataset → Validation → Cleaning → Verification → Rebalancing → Balanced Dataset Validation → Augmentation-Ready Dataset

3. Dataset Cleaning Pipeline

The cleaning pipeline began by locating and extracting the current dataset and establishing a clear baseline of the files available for processing. The original dataset was retained separately from the cleaned dataset to ensure that the cleaning process remained traceable and reproducible.

The existing Data QA outputs were then reused, including the dataset manifest and duplicate reports. These reports provided the validation information required to identify records requiring further cleaning.

The cleaning process applied several quality-control rules.

3.1 Empty File Removal

Files with zero bytes or missing file-size information were identified as empty and marked for removal.

The validation results showed that there were **0 empty files** in the dataset.

3.2 Unreadable File Removal

Files that were not identified as readable were marked for removal. Unreadable records cannot reliably proceed through audio processing and therefore should not be included in the cleaned dataset.

The validation identified **22 unreadable files**.

3.3 Short Recording Removal

Recordings shorter than one second were identified and marked for removal. This ensured that the dataset followed the defined minimum-duration requirement.

A total of **722 recordings were identified as being shorter than one second**.

3.4 Unsupported Format Removal

The cleaning pipeline retained the supported audio formats and marked records with unsupported extensions for removal.

The supported formats were:

- WAV
- MP3
- FLAC
- OGG

The validation identified **22 files with unsupported formats**.

3.5 Species Label Standardisation

Species labels were standardised to ensure that equivalent labels were treated consistently during subsequent dataset processing and balancing.

The standardisation process removed unnecessary whitespace, converted underscores into spaces, and consolidated multiple spaces. Original labels were retained so that changes could be traced.

This step was important because inconsistent species labels could result in the same species being treated as multiple classes during rebalancing.

3.6 Duplicate Removal

The duplicate reports generated during the Data QA process were reused to identify exact duplicate recordings across dataset buckets.

Where an exact duplicate was identified between Bucket 1 and Bucket 3, the Bucket 1 copy was retained and the Bucket 3 copy was marked for removal.

The validation identified **5 Bucket 3 duplicate records** for removal. Of these categories, **2 records also met the short-duration criterion**, demonstrating why overlap checks were required when calculating the final removal count.

4. Cleaning Verification

After applying the cleaning rules, additional verification checks were performed to ensure that records identified as problematic had not accidentally remained in the retained dataset.

The verification checks confirmed that:

- Short recordings still being kept: **0**
- Unsupported files still being kept: **0**
- Bucket 3 duplicates still being kept: **0**

These results indicate that the identified records were correctly handled by the cleaning process.

The cleaning manifest contained **15,574 records** before the cleaning decisions were applied.

The validation results were:

Cleaning Category	Records Identifie
Under 1 second	722
Unsupported format	22
Empty files	0
Bucket 3 exact duplicates(+2 records in both shorter than 1 sec and duplicates	3
Records marked for removal	747

The total number marked for removal was lower than the sum of all individual categories because some records satisfied more than one cleaning criterion. In particular, **2 records were both shorter than one second and Bucket 3 duplicates**.

This overlap analysis provided an additional consistency check and confirmed that the removal count was not being incorrectly inflated by counting the same record multiple times.

5. Cleaned Dataset Creation

Following the cleaning decisions, the retained records were copied into a separate cleaned_dataset directory.

The cleaned dataset was organised according to the standardised species labels, creating a consistent directory structure for the subsequent rebalancing stage.

A verification was also performed to ensure that every record retained in the final manifest was successfully copied into the cleaned dataset.

This separation between the original and cleaned datasets provides traceability and ensures that the original data remains available for reference while the cleaned version is used for subsequent processing.

6. Cleaning Reports and Documentation

The cleaning pipeline generated a set of reports to document the cleaning process and its outcomes.

These included:

- **Cleaning manifest** — records the cleaning information and decisions for the dataset.
- **Final cleaned manifest** — identifies the records retained after cleaning.
- **Removed files report** — records files marked for removal and their associated cleaning reasons.
- **Removal summary** — provides counts of files removed for each cleaning reason.
- **Species distribution after cleaning** — records the number of files available for each species following cleaning.

These outputs provide an auditable record of the cleaning process and support reproducibility.

7. Balanced Dataset Results

Following the cleaning stage, the dataset was rebalanced to produce a more consistent species distribution.

The final balanced dataset contained:

10,228 audio files

The species distribution was changed through the balancing process so that the classes were more evenly represented compared with the distribution of the cleaned dataset.

During the validation of the balanced dataset, **237 recordings were identified as too long**.

The presence of these 237 too-long recordings provides an important validation finding. It demonstrates that the final balanced dataset was independently checked rather than assuming that balancing alone guaranteed dataset validity.

The balanced dataset therefore provides the required foundation for the next processing stage, while the validation results highlight records requiring consideration before or during augmentation.

8. Sprint 2 Outputs

The Sprint 2 work addressed the required outputs as follows:

Cleaned and Validated Main Dataset

A cleaned dataset was created after applying the defined quality-control rules. Invalid, unreadable, duplicate, unsupported, and unsuitable records were identified and handled through the cleaning manifest.

Balanced-Dataset Validation Report

Validation was performed on the resulting balanced dataset, including assessment of the final audio records and identification of recordings exceeding the permitted duration.

The balanced dataset contained **10,228 audio files**, with **237 recordings identified as too long**.

Balanced Dataset Ready for Augmentation

The rebalancing stage produced a dataset with a changed and more balanced species distribution, providing the basis for the subsequent augmentation stage.

The validation results should be considered when proceeding to augmentation, particularly the identified too-long recordings, so that only records meeting the final quality requirements are passed forward.

9. Reproducibility and Quality Assurance

A key aspect of the Sprint 2 work was maintaining a reproducible and traceable cleaning process.

The pipeline preserved the original dataset separately, reused the existing Data QA validation outputs, recorded cleaning actions and reasons in manifests, generated summary reports, and performed verification checks after creating the cleaned dataset.

This approach ensures that dataset changes are documented rather than being performed through manual deletion or modification. The resulting reports provide evidence of which records were removed, why they were removed, and how the dataset changed through the cleaning and balancing stages.

The verification of the balanced dataset also ensured that the final dataset was assessed independently before progressing to augmentation.

10. Contribution Summary

The Sprint 2 contribution focused on supporting the **dataset cleaning, validation, and rebalancing workflow** required to prepare the dataset for augmentation.

The main contributions were:

- Established and applied the dataset cleaning pipeline to the current main dataset.
- Reused the existing Data QA validation and duplicate reports from the earlier dataset-quality process.
- Identified and handled unreadable, unsupported, short, empty, and duplicate records.
- Standardised species labels to improve consistency for class-based processing and balancing.
- Performed overlap checks to verify that removal counts were accurate.
- Created a separate cleaned dataset while preserving the original dataset.
- Verified that retained records were successfully transferred into the cleaned dataset.
- Generated cleaning manifests and summary reports to document the cleaning process.
- Supported the rebalancing process and assessed the resulting species distribution.
- Verified the balanced dataset, which contained **10,228 audio files**.
- Identified **237 too-long recordings** during balanced-dataset validation.
- Provided a documented and reproducible dataset-cleaning workflow that supports the transition to the augmentation stage.

Overall, the Sprint 2 work established a structured quality-control process from the original dataset through cleaning and rebalancing, providing a documented foundation for the next stage of dataset augmentation.

Final balanced data splits summary

audio files distribution (79 species in each set same species no difference) train - 6258

test - 4613

validation - 4466

duplicates (within each set)

train - 104 - within each species not mixed

exact duplicate summary in file_duplicates report in train_reports.zip

test - 0

validation - 0